

AI water consumption is negligible

Posted by u/GapTerrible2179 • 0 points • 10 comments

I see claims all the time that AI is using up extreme amounts of water and is completely unnecessary, so I thought I'd look into how much water is actually used. These calculations are for my own computer running a local model so my total water consumption is likely lower than the average given that people use ChatGPT and other non-local models, but on a person-by-person basis I can't imagine it's much of a difference.

My computer uses approximately 450 watts to generate an image at 1024x1024 resolution running SDXL 1.0 (a popular model, not the best or newest but still decent). It takes less than 30 seconds to generate an image, but for the sake of easy math we'll pretend it takes 30 seconds each time.

$$450 \text{ W} \times .0083 \text{ hours} = .00375 \text{ kWh}$$

According to Google there's a lot of variation in how much water is used in power plants for cooling (since my setup is local there's no data center cooling to take into account), but it seems much more consistent for data centers at an average of 1.8 L/kWh, and that's what most people would be using so I'll use that number.

$$1.8 \text{ L/kWh} \times .00375 \text{ kWh} = 6.75 \text{ mL}$$

So on average a single image generation takes less than 7 mL of water. But I can hear the counterargument already, that I'm not considering the training of the model, so let's factor that in as well.

Unfortunately there isn't readily available info that I could find for how many people use a given model, so we'll approximate it. Civitai, a popular platform for both downloading models and also generating images, has roughly 6 million monthly users (not visits, there are over 20 million monthly visits) from what info I could find online. Not everyone uses the same model but SDXL was pretty popular, so let's guesstimate and say 1 million of those 6 million users generated images using SDXL at least once. Obviously not everyone using SDXL is using Civitai, so the number of total people who have used SDXL is likely higher than 1 million, but we'll use 1 million to make the math simple.

There also isn't readily available info I could find for how much water was used to train SDXL, but I did see an estimate from a paper this year stating that training an LLM uses

approximately 2.769 million liters of water total, so let's use that number.

$$2.769 \text{ million L} / 1 \text{ million users} = 2.769 \text{ L/user}$$

Keep in mind that this is probably a much lower estimate of SDXL users than there actually are worldwide, so at most each person would be personally responsible for wasting about 3 liters of water one single time.

So all in all, even if you account for the water used during training the average user is responsible for about 3 liters of water after generating their first image, and 7 mL of water every image after that.

Again I can already hear the argument that AI is wholly unnecessary and all that water is truly wasted. Let's assume that's true, and compare it to another wholly unnecessary activity: golf. The average 18 hole golf course uses 1-2 million liters of water per week. That adds up to tens of millions of liters of water every year per golf course, all AI water consumption is a literal drop in the bucket comparatively.

Golf is known for wasting water though, so let's compare it to another activity, this time more necessary: showering. The average shower lasts about 8 minutes according to Google and uses 65.1 liters of water, so 1 minute showering uses about 8.1375 liters of water.

$$8.1375 \text{ L} - 3 \text{ L (the initial personal cost from training the model)} = 5.1375 \text{ L}$$

$$5.1375 \text{ L} / 7 \text{ mL} = 733.9$$

So for every minute I spend in the shower that isn't strictly necessary, I've wasted the same amount of water as if I had generated over 700 images on my computer at the low end. The actual number is likely in the thousands.

I have no choice but to conclude that maybe AI is not wasting as much water as is commonly assumed.

Comments

u/IAmNotATimepiece • 1 points • 2025-09-05 01:22 UTC

I guess everyone that's losing access to municipal water and having their lives ruined by data centers is lying. Good to know!

u/Even-Mode7243 • 1 points • 2025-09-03 23:20 UTC

I think the issue is not with AI (although AI is part of the equation) but with data centers. The video of the family that had a data center built 300ft from their house shows how obstructive they can be. Their water pressure is basically gone and the lights from the data center blast through their blinds at all hours. It doesn't matter if they are 100% environmentally neutral if they ruin the lives of people that live nearby.

u/soulmagic123 • 0 points • 2025-09-03 15:41 UTC

Your argument is completely insincere if it doesn't include model creation.

It's like "look over here, this is 12 paragraphs about the part of la that doesn't consume water"

Also we all agree the infrastructure is already over taxed.

Do you get to move into an apartment people are already living in?

I guess you do if you have trillions of dollars, but also why not use some of the money to fund desalination?

Old paper mills don't have that money.

The meat industry has fairly slim margins.

Golf courses close all the time.

But you, you have people falling all over themselves to give you huge piles of cash, use some of that cash to further desalination tech. It's not too much to ask, not even close.

u/GapTerrible2179 • 3 points • 2025-09-03 16:08 UTC

Training is much more difficult than creating the model and requires more resources, so I can't imagine the numbers would change all that much. It's hard to get reliable data because energy and water usage varies depending on the model and when and how it was trained, but estimates I've seen online say the training portion uses between about 50% to 90% of the total water used, with the rest being used during creation. I believe the 2.769 million liter figure I used took model creation into account as well, but even if it didn't and we assume the model creation was responsible for 50% of the water used we just need to double the end figure, so each person would be responsible for 5.538 liters total. That's the equivalent of showering for an extra 41 seconds one single time.

I do agree that AI companies should be funding more efficient and environmental solutions though, better desalination tech and more of it would be great. Personally I'd like to see a lot of changes to energy infrastructure, regardless of any impact from AI

u/soulmagic123 • 0 points • 2025-09-03 16:26 UTC

If it's 50 to 90 percent then you should probably mention that part in your argument. I'm not here to certify your numbers but I will say you can already find real world use case of counties losing most of their fresh water to data centers because they didn't read they fine print (Google Arizona) There's a reason these companies tend to pop up in poorer places that don't have the money to fight back or even say no to this type of business infrastructure.

These ai companies are all following the standard tech model , lose money until you kill the competition then profit. It's the same thing as Amazon with brick and mortar or uber with taxis.

In this case it's real writers, artist, content creators, etc, it's lose money until these real

people are dead, then jack up the prices and profit.

If you allowed to lose money then you should lose some of that money turning salt water into fresh water , or is that too "actually not an evil thing" for these companies to adopt?

In other words wouldn't be nice if we got something good at this deal, like solving huge problem the will probably be the source of the future wars?
Or is it just take take take?

u/GapTerrible2179 • 1 points • 2025-09-03 17:14 UTC

I agree that was an oversight on my part. I was using the upper bound of water usage for most of my calculations and the estimates I found online are all likely higher than the actual water usage of image generation models in particular, so I believe that even counting model creation it's unlikely each person would be responsible for even 5 liters total but you're right that I should've included it in the calculations.

As for communities losing their water that seems like less of an AI problem to me and more of a capitalism problem. AI as a percentage of data center usage is between 10% and 24% from what I saw during a quick Google search, so these communities would be losing water even if AI disappeared tomorrow. I think there needs to be stronger regulation on the entire industry but it's not an AI-specific problem, and AI may actually help via optimization.

I agree completely that AI companies should be using more of their money to make the world better and fix a lot of these problems, but unfortunately they do not ask me for my opinion

u/raiderwolf • 1 points • 2025-09-03 17:13 UTC

They did mention that in their argument, they discuss training figures and how even considering those it's still not much water.

The local water issue is a problem for that local area for sure, because its using one tiny locations water to provide a service to the whole world basically. That needs regulation and laws to locate the data centers in places with sufficient water that won't impact local population. But in global terms it is a drop in the bucket is the point they are making I think

u/Similar_Geologist_73 • 0 points • 2025-09-03 06:15 UTC

If it's negligible, then why do yall keep bringing it up

u/Kolbr00 • 5 points • 2025-09-02 22:18 UTC

This is a very thoughtful and well calculated assessment. Now that I think about it, the argument of water consumption has already been placed on the table. I also think that rather than using these issues as a reason of why something isn't good for usage, we use this problem as the foot of our question of how do we resolve this issue to better the environmental health and advance AI. It was widely known this age is to come, so we should use that as a means to better ourselves and what we love.

These calculations are well thought, so I say good!

u/ResourceFront1708 • 7 points • 2025-09-02 22:14 UTC

Actually educated antis know this

u/redditscraperbot2 • 7 points • 2025-09-02 23:29 UTC

The question is what they do with that knowledge.